



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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LAB_MST

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Section: 23AIT KRG 1(G2)

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1. **Aim:** : To create a simple React application demonstrating protected routing using the Context API for global authentication state. The application will have a login page to authenticate the user and a protected dashboard page that is accessible only after successful login.

2. Implementation/Code

Login.js:

```
import { useState, useContext } from "react";
import { AuthContext } from "../context/AuthContext";
import { useNavigate } from "react-router-dom";

function Login() {

  const [id, setId] = useState("");
  const [password, setPassword] = useState("");

  const { login } = useContext(AuthContext);

  const navigate = useNavigate();

  const handleLogin = () => {

    const success = login(id, password);

    if (success) {
      navigate("/dashboard");
    } else {
      alert("Invalid ID or Password");
    }
  }
}
```

```

    }
  };

  return (

    <div
      style={{
        backgroundColor: "#e3f2fd",
        height: "100vh",
        display: "flex",
        justifyContent: "center",
        alignItems: "center"
      }}
    >

      <div
        style={{
          backgroundColor: "white",
          padding: "40px",
          borderRadius: "10px",
          boxShadow: "0px 0px 10px gray",
          textAlign: "center"
        }}
      >

        <h2>Login Page</h2>

        <input
          type="text"
          placeholder="Enter ID"
          value={id}
          onChange={ (e) => setId(e.target.value) }
        />

        <br /><br />

        <input
          type="password"
          placeholder="Enter Password"
          value={password}
          onChange={ (e) => setPassword(e.target.value) }
        />

        <br /><br />

        <button onClick={handleLogin}>Login</button>

```

```

        </div>

    </div>
  );
}

export default Login;

```

Dashboard.js:

```

function Dashboard() {

  return (
    <div>
      <h1>Welcome to EcoTrack Dashboard</h1>
    </div>
  );
}

export default Dashboard;

```

ProtectedRoute.js:

```

import { useContext } from "react";
import { Navigate } from "react-router-dom";
import { AuthContext } from "../context/AuthContext";

function ProtectedRoute({ children }) {

  const { isAuthenticated } = useContext(AuthContext);

  if (!isAuthenticated) {
    return <Navigate to="/login" />;
  }

  return children;
}

export default ProtectedRoute;

```

AuthContext.js:

```
import { createContext, useState } from "react";

export const AuthContext = createContext();

export const AuthProvider = ({ children }) => {

  const [isAuthenticated, setIsAuthenticated] = useState(false);

  const login = (id, password) => {

    if (id === "admin" && password === "1234") {
      setIsAuthenticated(true);
      return true;
    }

    return false;
  };

  return (
    <AuthContext.Provider value={{ isAuthenticated, login }}>
      {children}
    </AuthContext.Provider>
  );
};
```

Index.js:

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";

const root = ReactDOM.createRoot(document.getElementById("root"));

root.render(
  <App />
);
```

App.js:

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Login from "../pages/Login";
import Dashboard from "../pages/Dashboard";
import ProtectedRoute from "../components/ProtectedRoute";
import { AuthProvider } from "../context/AuthContext";

function App() {
  return (
    <AuthProvider>
      <BrowserRouter>
        <Routes>

          <Route path="/" element={<Login />} />

          <Route path="/login" element={<Login />} />

          <Route
            path="/dashboard"

            element={

              <ProtectedRoute>

                <Dashboard />

              </ProtectedRoute>
            }
          />

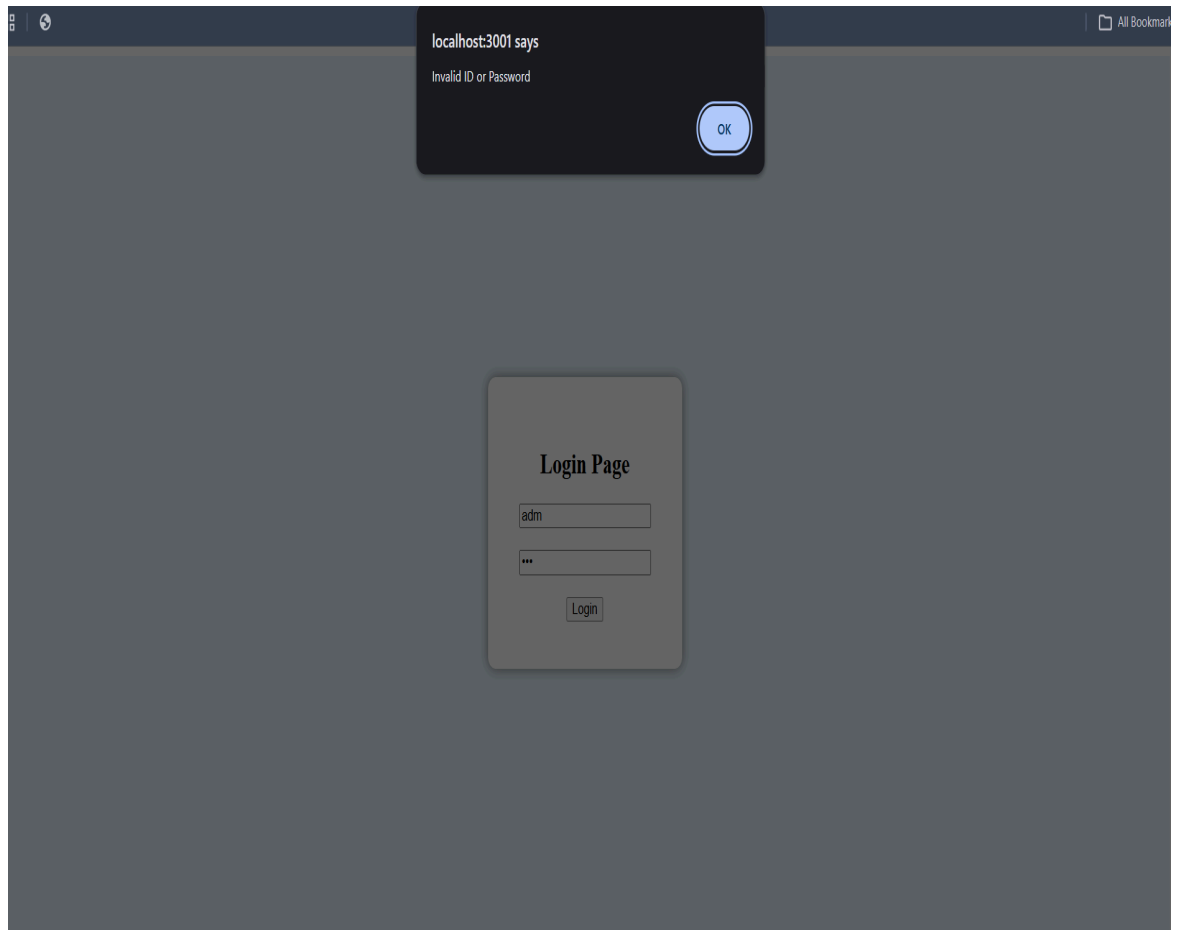
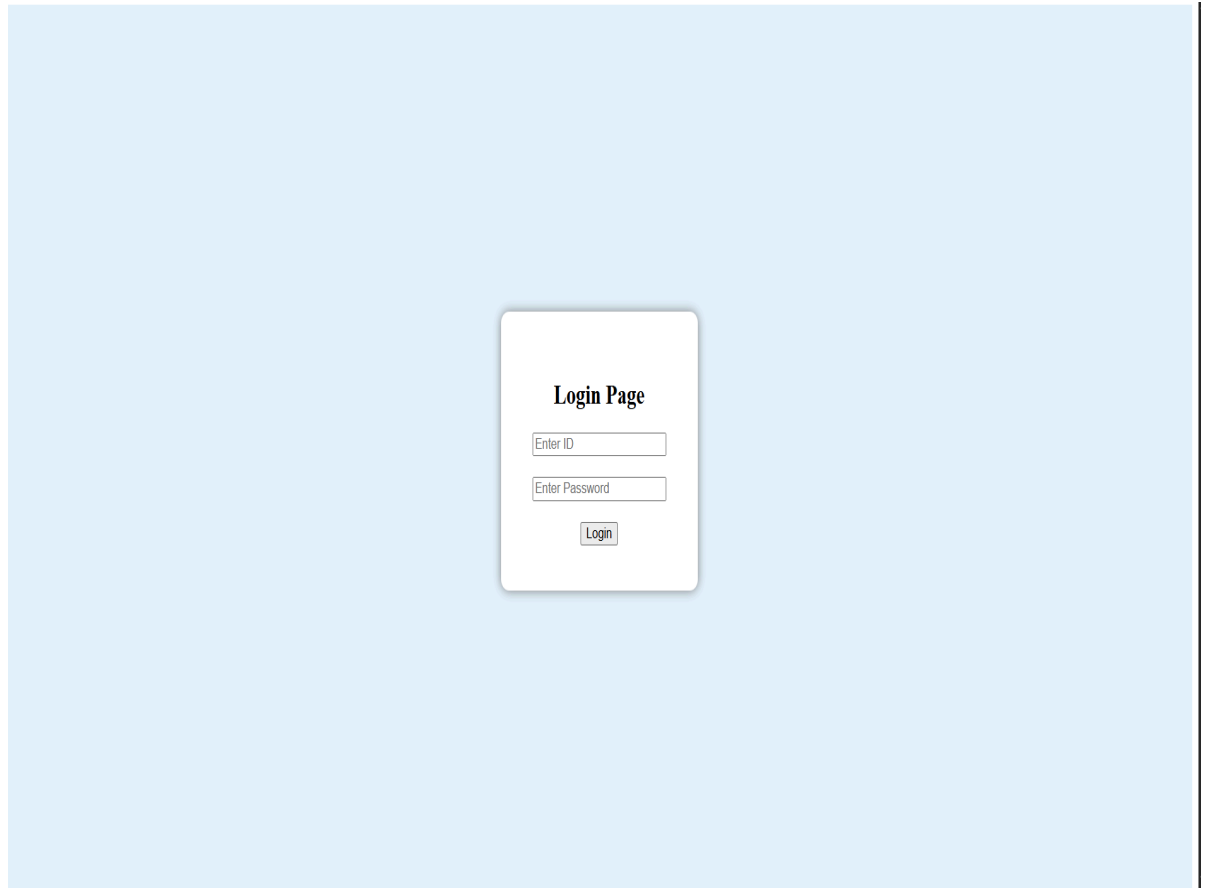
        </Routes>

      </BrowserRouter>
    </AuthProvider>
  );
}

export default App;
```

3. Output:

Login page:



Welcome to EcoTrack Dashboard

